TruthLens AI
AI-Powered Cross-Document Consistency & Forgery Detection System for Banking Underwriting

Hackathon: Canara Bank SuRaksha Cyber Hackathon 2.0
Theme: Real-Time Anomaly Detection in Banking

CONTEXT — Integration Context

Banks rely heavily on customer-submitted documents during underwriting, including land records, bank statements, GST filings, financial statements, legal agreements, and collateral documents. These records are often submitted in scanned or digital form, making them vulnerable to tampering, forgery, and hidden inconsistencies. Current verification processes are largely manual and fragmented. Verification teams validate documents individually, creating gaps where fraudsters may manipulate one document while failing to maintain consistency across the complete application. TruthLens AI acts as an intelligent underwriting verification layer that automatically compares information across multiple submitted documents in real time. Instead of checking whether a single document appears authentic, the system validates whether all submitted records collectively form a logically consistent and trustworthy financial profile.

RELEVANCE — Theme Alignment

The solution directly aligns with the theme of Real-Time Anomaly Detection by identifying suspicious inconsistencies, forged modifications, and financial anomalies before loan approval. Unlike traditional systems that rely only on OCR extraction or isolated document checks, TruthLens AI focuses on cross-document intelligence and cybersecurity-backed verification. The platform improves: • Fraud prevention during underwriting
• Faster and more reliable loan processing
• Reduced manual verification effort
• Better compliance and auditability
• Real-time anomaly-based risk assessment

SOLUTION OVERVIEW — TruthLens AI

TruthLens AI is an AI-powered underwriting intelligence platform that detects forgery attempts, metadata tampering, and logical inconsistencies across multiple financial and legal documents. The core innovation is the

Cross-Document Consistency Engine, which verifies whether all submitted documents collectively represent a believable and legally valid financial reality. The platform combines: • OCR-based data extraction

• Cross-document entity comparison
• Metadata tampering checks
• Financial anomaly analysis
• Explainable fraud insights
• Cybersecurity-backed integrity verification The system generates: • Fraud risk scores
• Contradiction reports
• Underwriting recommendations
• Confidence indicators

KEY FEATURES

1. Multi-Format Document Ingestion

Supports PDFs, scanned documents, bank statements, land records, GST filings, legal agreements, and financial statements through secure upload APIs.

2. Cross-Document Consistency Engine (Core Innovation)

Extracts and compares: • Names

• Addresses
• PAN / Aadhaar identifiers
• Property details
• Ownership timelines
• Financial values Detects: • Mismatched ownership records
• Contradictory timelines
• Inflated valuations
• Inconsistent financial claims

3. Forgery & Metadata Detection

Detects: • Modified PDF metadata

• Font inconsistencies
• Edited scans
• Copy-paste tampering
• Suspicious timestamps

4. Financial Behavioral Validation

Analyzes transaction patterns, turnover consistency, and unusual financial spikes to identify suspicious business activity.

5. Cybersecurity Integrity Layer

Implements: • SHA-256 document fingerprinting

• Secure encrypted uploads

• Role-based access control

• Audit logging for traceability

• Tamper alerts on document re-submission

6. Explainable AI Dashboard

Provides: • Risk scores

• Contradiction matrix

• Highlighted inconsistencies

• Fraud probability indicators

• Human-readable explanations

MODEL ARCHITECTURE

Architecture Type:

Hybrid document intelligence + semantic consistency validation + anomaly detection framework

Core Components:

• OCR extraction engine

• NLP-based entity extraction

• Metadata tampering analysis

• Cross-document consistency validation

• Financial anomaly detection

• Risk scoring engine

Detection Layer:

• Rule-based validation

• Isolation Forest anomaly detection

• Semantic similarity comparison

• Metadata integrity analysis

TECHNICAL STACK

Frontend: React Dashboard

Backend: Python FastAPI

OCR: Tesseract OCR

NLP: spaCy + Sentence Transformers

Computer Vision: OpenCV

Database: PostgreSQL

Authentication: JWT + RBAC

INNOVATION HIGHLIGHTS

• Cross-document consistency intelligence instead of isolated document checks

• Contradiction Matrix for explainable anomaly visualization

• Cybersecurity-backed tamper verification using document fingerprinting

• Real-time underwriting anomaly detection

• Financial behavior validation instead of only static document checks

• Explainable AI insights for human underwriters

HACKATHON MVP SCOPE

The hackathon prototype will focus on: • Uploading multiple underwriting documents

• OCR extraction and entity identification

• Cross-document comparison and contradiction detection

• Metadata tampering checks

• Risk scoring dashboard

• Secure upload + document fingerprinting layer **Example Fraud Scenarios Demonstrated:**

• Land ownership mismatch across records

• GST turnover inconsistent with bank statement activity

• Edited PDF metadata detection

• Inflated collateral valuation

SUCCESS METRICS

• Reduction in manual verification effort

• Faster underwriting decision-making

• Fraud detection accuracy

• Cross-document consistency accuracy

• Reduced approval of high-risk applications

• Average processing time per application

FUTURE IMPROVEMENTS

The following enterprise-scale features are planned as future enhancements beyond the hackathon MVP: • Blockchain-based immutable document registry

• Graph-based fraud ring detection using Neo4j

• Federated learning across bank branches

• SIEM and SOC integration for enterprise monitoring

• Advanced adversarial AI testing

• Multi-region cloud deployment

• Integration with MCA21, CERSAI, and CIBIL APIs

• Indic language expansion for rural banking workflows

CONCLUSION

TruthLens AI transforms underwriting from isolated document verification into intelligent consistency-based trust validation. By combining AI-driven anomaly detection with cybersecurity-backed integrity verification, the platform helps banks detect fraud earlier, reduce operational overhead, and improve the reliability of underwriting decisions in real time. **TruthLens AI — Where documents are not only verified, but logically validated.**
